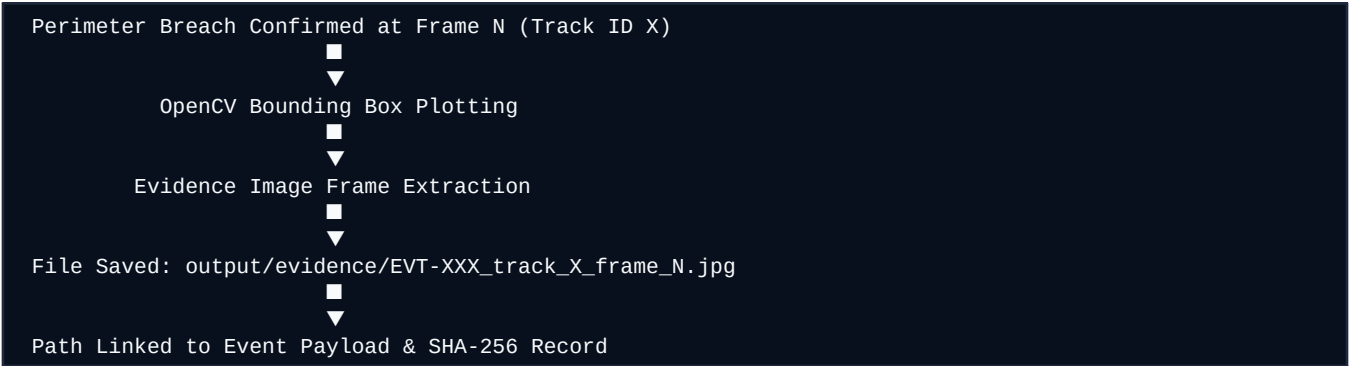


IBVAP — Digital Evidence Capture & Storage

07. Visual Evidence Management & Immutable Association

1. Evidence Snapshot Generation Workflow

When the intrusion detection engine confirms a perimeter breach at frame \$N\$, the video processing pipeline executes an immediate frame grab:



2. Evidence Naming Convention & Directory Mapping

Project Evidence Files (Verified from Source)

Event ID	Track ID	Frame	File Name	Disk Path
EVT-001	Track 2	Frame 30	EVT-001_track_2_frame_30.jpg	output/evidence/EVT-001_track_2_frame_30.jpg
EVT-002	Track 7	Frame 108	EVT-002_track_7_frame_108.jpg	output/evidence/EVT-002_track_7_frame_108.jpg
EVT-003	Track 36	Frame 177	EVT-003_track_36_frame_177.jpg	output/evidence/EVT-003_track_36_frame_177.jpg

Additional Evidence Repositories in Codebase:

- `src/output/evidence/evidence_001.jpg, evidence_002.jpg, evidence_003.jpg` (Secondary fallback repository)
- `screenshots/intrusion_001.jpg, intrusion_002.jpg, intrusion_003.jpg` (UI demonstration snapshots)

3. Streamlit Evidence Gallery & High-Res Viewer

In the Streamlit Dashboard (`streamlit_app/pages/digital_evidence.py` & `src/app.py` Lines 1009–1080):

- **Dynamic Image Fetcher:** `fetch_image_bytes(source)` dynamically loads images from local disk paths or remote URLs with timeout and memory caching (`@st.cache_data(ttl=300)`).
 - **Interactive Cards:** Each evidence snapshot is rendered alongside Event ID, Track ID, Frame number, Timestamp, and Risk Badge.
 - **High-Resolution Zoom Viewer:** Clicking "Zoom" opens a modal overlay displaying the raw full-resolution evidence image for detailed security inspection.
 - **Direct Download:** Users can download evidence snapshots (`.jpg`) directly from the UI for forensic handoff.
-

4. Digital Evidence Integrity & Immutability

- **SHA-256 Hash Association:** The evidence filename and metadata are included in the JSON string payload that generates the SHA-256 hash record in `secure_audit_log.csv`.
- **Tamper Evidence:** If an attacker modifies or replaces an evidence image file on disk, the associated audit hash validation will fail upon root digest verification.
- **Future Immutable Evidence Scope:** Storing evidence SHA-256 hashes on an IPFS (InterPlanetary File System) decentralized network or permissioned blockchain smart contract for legal admissibility.